

Test 1

EE112FZ, Semester 2, 2025 Spring (APRIL 15)

FZU Number: _____ Name: _____ Score: _____

1. [10 marks] Find the Laplace transform of $f(t) = 2 \sin(5t + 4)$.
2. [10 marks] Find Laplace transform of $f(t) = (t - 1)^2 U(t - 5)$.
3. [10 marks] Find the inverse Laplace transform of $F(s) = \frac{s + 1}{(s + 1)^2 + 16}$.
4. [10 marks] Find the inverse Laplace transform of $F(s) = \frac{s + 1}{(s - 3)^2}$.
5. [10 marks] Using Laplace transform and inverse Laplace transform to solve the equation
$$y''(t) + 5y'(t) + 6y(t) = 0, \quad y(0) = 1, \quad y'(0) = 0.$$
6. [10 marks] Let $\mathbf{A} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $\mathbf{B} = 2\mathbf{i} - \mathbf{j} + \mathbf{k}$ and $\mathbf{C} = 3\mathbf{j} + 2\mathbf{k}$. Find $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})$.
7. [10 marks] Let $\mathbf{A} = \mathbf{i} + \mathbf{j}$, $\mathbf{B} = \mathbf{i} - \mathbf{j}$. Find $\mathbf{A} \times (\mathbf{A} \times \mathbf{B})$.
8. [10 marks] Find the intersecting point of the straight line $\frac{x - 1}{2} = \frac{y}{2} = z$ and the
plane $x - y + z = 0$.
9. [10 marks] Find the symmetric equation of the line which passes through the point $\mathbf{P}(1, 1, 1)$
and is orthogonal to the plane $x + y + z = 1$.
10. [10 marks] The equation of the curve is given by $\mathbf{r}(t) = t\mathbf{j} + \cos t\mathbf{k}$. Find the unit tangent
vector $\hat{\mathbf{u}}$, the unit principal normal vector $\hat{\mathbf{N}}$ and the curvature κ at $t = 0$.